

COURSE CODE: CSE 4204 || COURSE TITLE: MOBILE COMPUTING LAB

**WEEK 06 TASK: BACKEND DEVELOPMENT & DATABASE
IMPLEMENTATION PROGRESS REPORT**

TEAM NAME: CSE4204-8D-T01 || SECTION: 8D

Project Title: NUBTK_COPILOT - AI-Powered University ERP and Career Architect

Instructor: Md. Riaz Mahmud, Assistant Professor, Dept of CSE, NUBTK

Section: 8D || **Team ID:** CSE4204-8D-T01

GitHub Repository: <https://github.com/bashirnubtk/cse4204-8d-t01-nubtk-copilot>

1. Team Information & Core Engineering Roles

Full Member Name	Student ID	System Design & Engineering Responsibilities
Bashir Alam	11220320963	Team Leader. Core backend system architecture, API design, Celery worker workflow orchestration, Open Router API & ML models integration.
Tisha Rani Chakroborty	11220320954	Frontend Developer. UI/UX layout design systems, interactive dashboard wireframes, and responsive views matching via Tailwind CSS.
Fariha Rahman Oishee	11220320967	Database Manager. PostgreSQL schema structure planning, entity relationship mapping, and relational table indexing within Supabase.
Ratul Raha Shuvo	11220320949	Backend Support. Payment gateway integrations sub-system design, transactional report templates, and REST API test suite architecture.

2. Authentication & Authorization Workflow

To safeguard corporate academic identities, a highly secure JWT-based stateless framework was implemented:

1. **Registration:** The client applications send payloads containing the user credentials to the backend. The application layer interceptor hashes the cleartext passwords using secure PBKDF2 blocks before saving them to the user schema.
2. **Login Verification:** Upon sending active credentials to the server, the backend system evaluates the cryptographic hash values. If authenticated, the service issues an isolated, time-bound **Access Token** and a long-lived **Refresh Token**.
3. **Route Protection:** Protected API endpoints (e.g., logging skills or launching mentor chats) require a valid Bearer <Token> string embedded directly into the standard incoming HTTP Authorization requests Header.

Backend Folder Structure:

```

backend_root/
|
|— .env                # System Environment Variables (API Keys, DB Credentials)
|— db.sqlite3          # Local Development Database Instance
|— manage.py           # Django Management CLI Utility
|— requirements.txt     # Project Python Dependencies & Packages
|
|— backend/            # Core Project Configuration Directory
|   |— __init__.py
|   |— asgi.py          # Asynchronous Server Gateway Interface
|   |— settings.py      # Main Project Configurations & Installed Apps
|   |— urls.py          # Global URL Routing & Endpoint Mapping
|   |— wsgi.py          # Web Server Gateway Interface Configuration
|
|— portal/             # Main Architecture Application Module
|   |— __init__.py
|   |— admin.py         # Django Admin Panel Models Registration
|   |— apps.py          # Application Context Configuration
|   |— context_processors.py # Custom Global Template Context Injectors
|   |— models.py        # Database Schema Blueprint Definitions
|   |— tests.py         # Backend Automation Test Scripts
|   |— urls.py          # Portal-Specific Endpoint Routing Matrices
|   |— utils.py         # AI Inference Engine & Helper Utilities
|   |— views.py         # Core Controller Logic & Request Handlers
|
|— migrations/         # Database Schema Version Control Logs

```

```

| | — 0001_initial.py    # Automated Supabase/SQL Database Migrations
| |   — __init__.py
|
| — static/              # Static Asset Repositories
| | — css/
| |   — output.css      # Compiled Tailwind CSS Utility Layer
| |   — js/
| |     — main.js       # Client-Side Interactive Scripts
|
| — templates/           # Server-Side Rendered HTML Layer
|   — academic_matrix.html # Semester Course Mapping Interface
|   — admin_dashboard.html # Superuser Management Control Center
|   — base.html          # Main Global Blueprint Layout Shell
|   — chat.html          # Real-Time AI Copilot Interaction Console
|   — home.html          # Institutional ERP Entry Landing Page
|   — login.html         # Secure User Authentication Gateway
|   — student_profile.html # Dynamic Skill Ledger & Portfolio View

```

3.1 User Registration Testing (POST /api/auth/register/)

Home
Portal Login
Admission 2026

✓ Application submitted! Your Enrollment ID is NUBTK-2026-1859.

Session 2026 • CSE Department Enrollment

Personal Details

Student Full Name *

Father's Name *

Mother's Name *

Contact Information

Phone Number *

Email Address *

```

[27/Jun/2026 19:46:57] "GET /portal/admission/apply/ HTTP/1.1" 200 10121
[27/Jun/2026 19:49:51] "POST /portal/admission/apply/ HTTP/1.1" 302 0
[27/Jun/2026 19:49:51] "GET / HTTP/1.1" 200 18015

```

7.4 AI Mentor Chat Testing (POST /api/ai-mentor/chat/)

NUBTK COPILOT

HomeDashboardLogout

AI Mentor Core Memory

This live metadata vector is being injected into the LLM system prompt in real-time:

Bashir Alam

75% Ready

Data Science & Machine Learning

3.64

OpenRouter Gateway Active

NUBTK Copilot AI Workspace

Gemini 2.5 Engine

Hello Bashir Alam! I am your AI Mentor.

Ask me anything about your academic track, due bills, or career recommendations.

NUBTK Copilot

নুবটিকে হাইব্রিড সার্ভার ক্লাস্টারে অতিরিক্ত ট্রাফিকের কারণে রেসপন্স জেনারেট করা যায়নি। আবার চেষ্টা করুন।

Type your academic doubt or ask to review CV/Slide image here...

Send

ut_shape`/input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

super().__init__(activity_regularizer=activity_regularizer, **kwargs)

WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.

[27/Jun/2026 21:30:28] "GET /portal/student/dashboard/ HTTP/1.1" 200 12268

[27/Jun/2026 21:31:54] "GET /student-portal/practical-skills/ HTTP/1.1" 200 9009

[27/Jun/2026 21:32:43] "GET /student-portal/ai-chat-hub/ HTTP/1.1" 200 11067

[27/Jun/2026 21:32:48] "POST /student-portal/ai-chat-hub/ HTTP/1.1" 200 627